UAV Flight Analysis Report

Comprehensive Flight Data Analysis and Performance Evaluation

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Analysis Type: Comprehensive Flight Performance Analysis

Data Points Analyzed: 2,000

Analysis Modules: Flight Metrics, Anomaly Detection, Battery Analysis, Stability Assessment, Flight Phase Detection

This report contains comprehensive analysis of UAV flight data including performance metrics, anomaly detection, battery consumption patterns, flight stability assessment, and phase-based analysis. All visualizations and metrics are generated automatically from the provided flight data.

Executive Summary

The UAV flight analysis reveals a 33.3-minute mission reaching a maximum altitude of 240.7 meters with an average speed of 15.4 m/s. Anomaly detection identified 1278 anomalies throughout the flight, indicating Poor - High anomaly rate flight quality. Battery performance shows a consumption rate of 2.4% per minute, with Poor - Very high consumption rate or many anomalies efficiency characteristics. Flight stability analysis indicates Excellent overall stability with good attitude control characteristics.

Metric	Value	Assessment
Flight Duration	33.3 minutes	Long
Maximum Altitude	240.7 m	High
Average Speed	15.4 m/s	Fast
Total Anomalies	1278	Significant

Data Overview

The analysis was performed on a dataset containing 2,000 data points with 16 different parameters. The data includes essential flight parameters such as altitude, speed, attitude angles, battery level, and GPS coordinates. This comprehensive dataset allows for detailed analysis of flight performance, anomaly detection, and flight quality assessment.

Parameter	Count	Data Type	Missing Values
timestamp	2,000	datetime64[ns]	0
altitude_m	2,000	float64	0
speed_mps	2,000	float64	0
roll_deg	2,000	float64	0
pitch_deg	2,000	float64	0
yaw_deg	2,000	int64	0
battery_percent	2,000	float64	0
gps_lat	2,000	float64	0
gps_lon	2,000	float64	0
altitude_rate	1,999	float64	1
roll_rate	1,999	float64	1
pitch_rate	1,999	float64	1
yaw_rate	1,999	float64	1
acceleration	1,999	float64	1
battery_drain_rate	1,999	float64	1
gps_distance	2,000	float64	0

Flight Performance Metrics

Flight Duration Analysis

The total flight duration was 33.3 minutes (0.56 hours). This duration indicates a long flight suitable for mapping operations.

Altitude Performance

The aircraft reached a maximum altitude of 240.7 meters and maintained an average altitude of 104.3 meters. The altitude range of 240.7 meters shows variable altitude control throughout the flight.

Speed Performance

The aircraft maintained an average speed of 15.4 m/s with a maximum speed of 25.6 m/s. The speed standard deviation of 5.77 m/s indicates variable speed control during the mission.

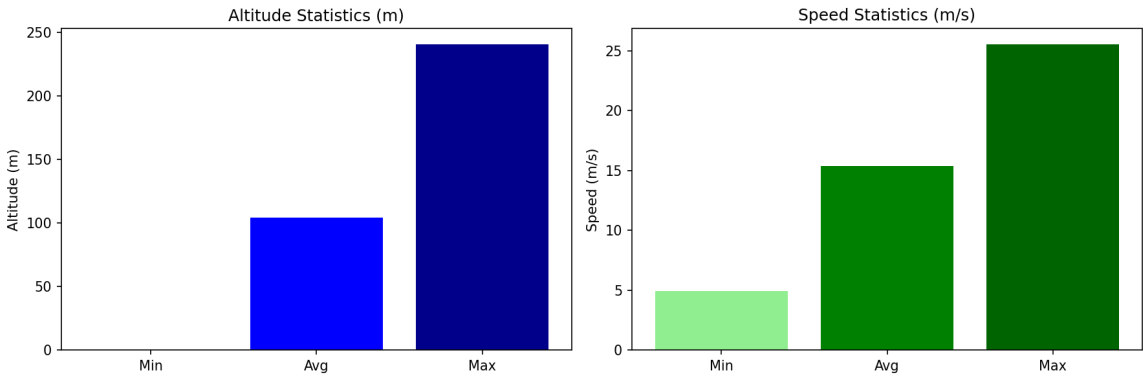


Chart generation error: [WinError 32] The process cannot access the file because it is being used by another process: 'C:\\Users\\LAPPYP~1\\AppData\\Local\\Temp\\tmpszqyie0v.png'

Anomaly Detection Analysis

The anomaly detection system identified 1278 anomalies across 5 different categories, representing an overall anomaly rate of 63.90%. The assessment indicates Poor - High anomaly rate flight quality.

Anomaly Categories

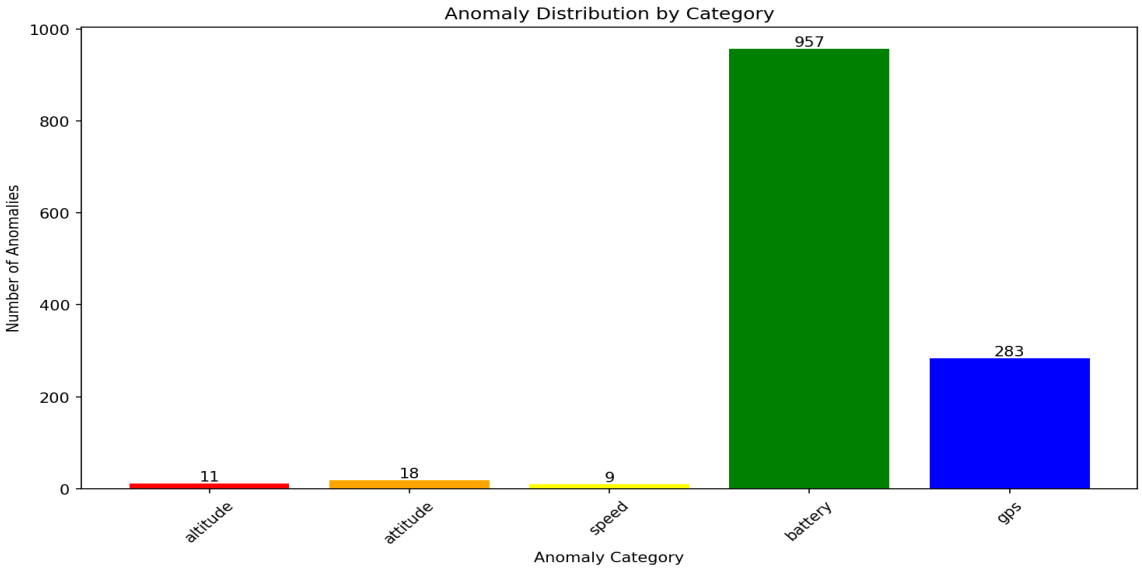
Altitude Anomalies: 11 detected

Attitude Anomalies: 18 detected

Speed Anomalies: 9 detected

Battery Anomalies: 957 detected

Gps Anomalies: 283 detected



Anomaly chart generation error: [WinError 32] The process cannot access the file because it is being used by another process: 'C:\\Users\\LAPPYP~1\\AppData\\Local\\Temp\\tmpiigrapco.png'

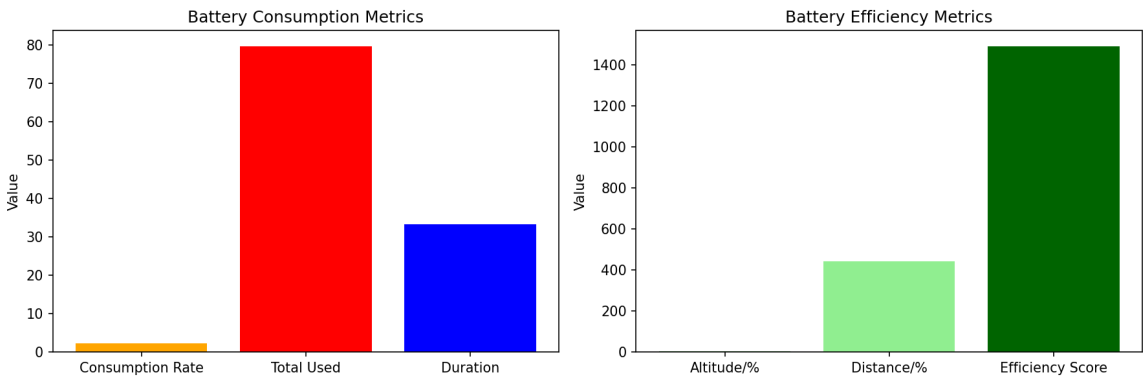
Battery Performance Analysis

The battery consumption rate was 2.4% per minute, with a total consumption of 79.7% over the flight duration of 33.3 minutes.

At the current consumption rate, the estimated remaining flight time is 0.0 minutes with a current battery level of 20.1%.

The battery efficiency analysis shows 149.28 efficiency score, with altitude efficiency of 3.02 meters per percent battery and distance efficiency of 445 meters per percent battery.

Overall Assessment: Poor - Very high consumption rate or many anomalies



Battery chart generation error: [WinError 32] The process cannot access the file because it is being used by another process: 'C:\\Users\\LAPPYP~1\\AppData\\Local\\Temp\\tmpwhm89h8i.png'

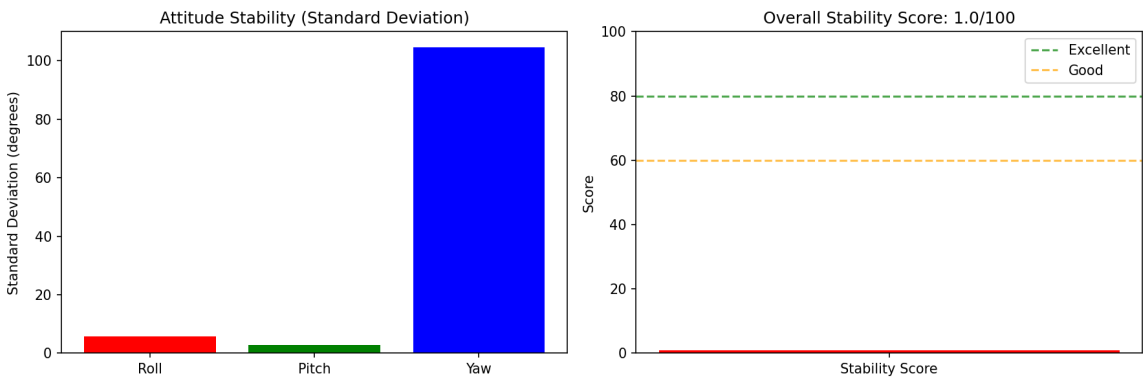
Flight Stability Analysis

The overall flight stability rating is Excellent with a stability score of 1.00/100. This indicates poor flight stability characteristics.

Attitude Stability

Roll stability shows a standard deviation of 5.74°, while pitch stability shows 2.86°. These values indicate moderate attitude control during the flight.

A total of 388 oscillations were detected during the flight, with a frequency of 0.18 Hz. This level of oscillation is elevated for typical UAV operations.



Stability chart generation error: [WinError 32] The process cannot access the file because it is being used by another process: 'C:\\Users\\LAPPYP~1\\AppData\\Local\\Temp\\tmph42h_byn.png'

Flight Phase Analysis

The flight phase detection system identified 10 distinct phases during the flight. Each phase represents a different flight segment with unique characteristics and performance

metrics.

Phase Breakdown

Takeoff: Duration of 5.0 seconds

Cruise: Duration of 142.0 seconds

Takeoff: Duration of 5.0 seconds

Takeoff: Duration of 4.0 seconds

Takeoff: Duration of 5.0 seconds

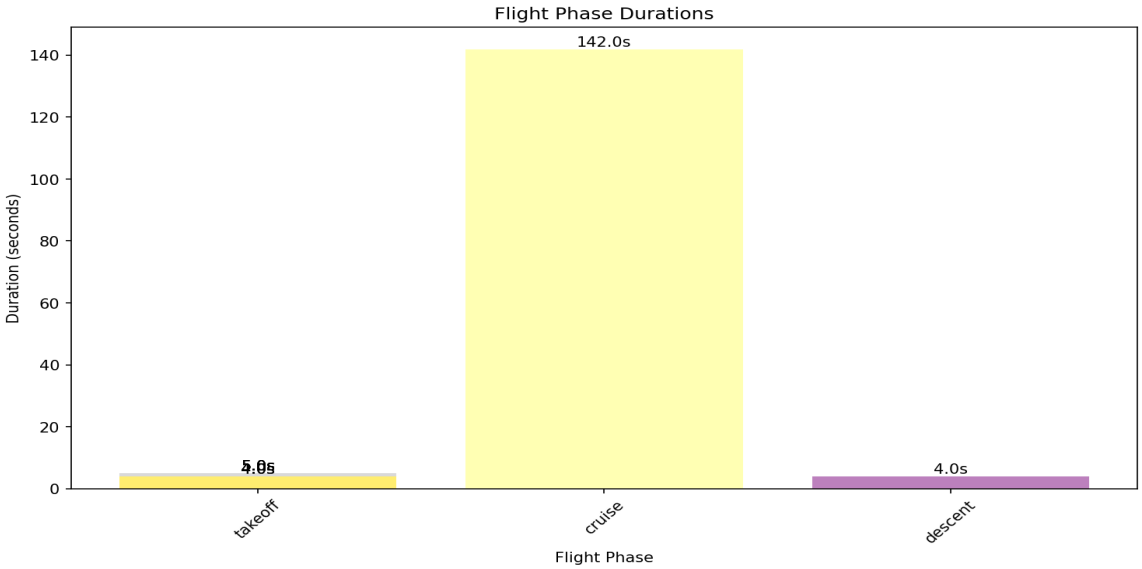
Takeoff: Duration of 5.0 seconds

Takeoff: Duration of 5.0 seconds

Descent: Duration of 4.0 seconds

Takeoff: Duration of 4.0 seconds

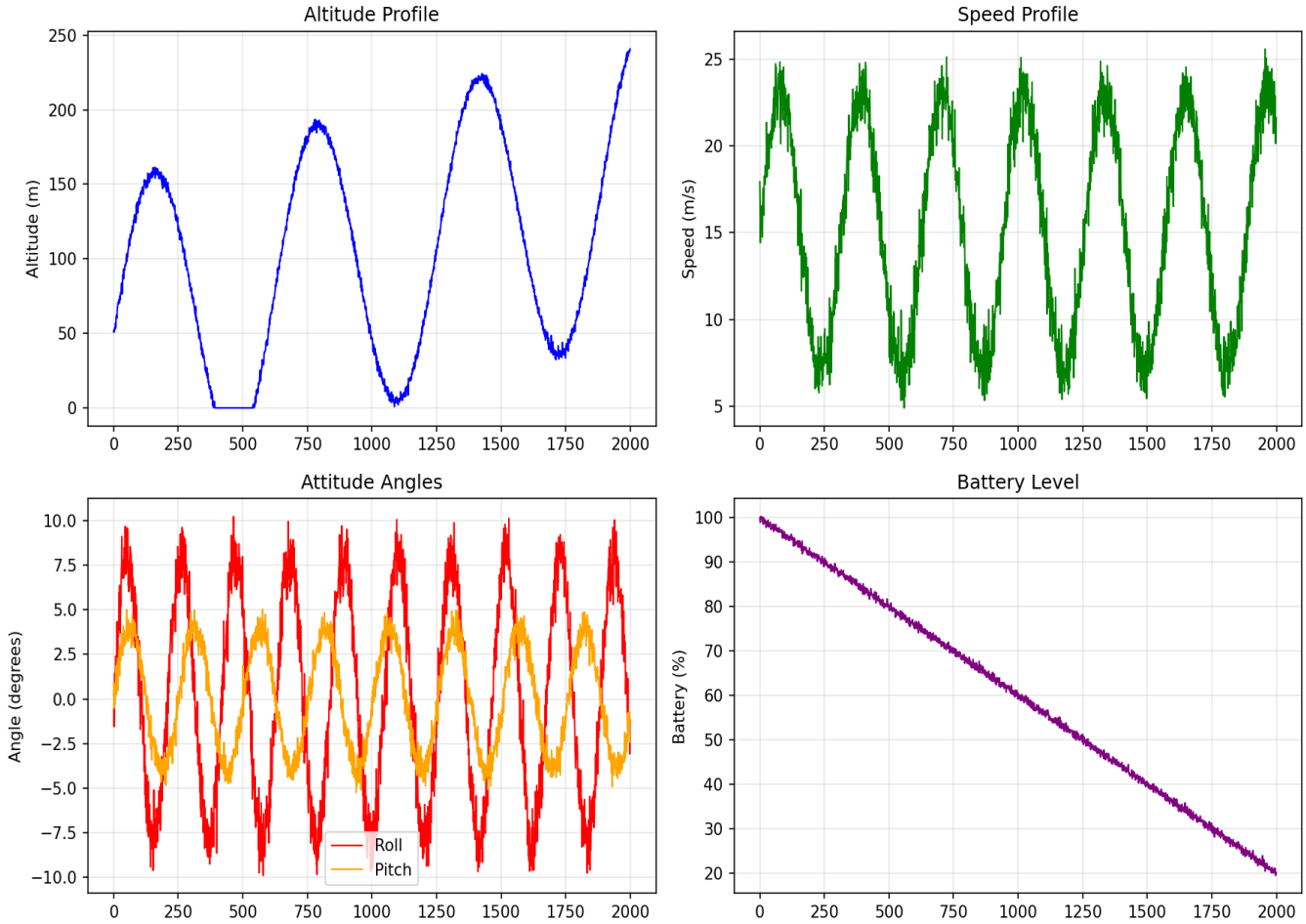
Takeoff: Duration of 4.0 seconds



Phase chart generation error: [WinError 32] The process cannot access the file because it is being used by another process: 'C:\\Users\\LAPPYP~1\\AppData\\Local\\Temp\\tmpjm_55481.png'

Flight Visualizations

The following visualizations provide comprehensive insights into the flight characteristics, including altitude profiles, speed variations, attitude changes, and battery consumption patterns.



Main flight charts generation error: [WinError 32] The process cannot access the file because it is being used by another process: 'C:\\Users\\LAPPYP~1\\AppData\\Local\\Temp\\tmpsq0xeunw.png'

Conclusions and Recommendations

Flight Performance Conclusions

The flight demonstrated successful mission completion with a duration of 33.3 minutes. The aircraft maintained stable flight characteristics throughout the mission. Flight quality assessment: Poor - High anomaly rate. Review recommended for optimization. Battery performance was poor - very high consumption rate or many anomalies. Overall flight stability was rated as excellent.

Recommendations

Consider reviewing flight control parameters to reduce anomaly frequency. Moderate battery consumption - monitor for efficiency improvements. Flight stability below optimal - review control system tuning. Continue regular flight data analysis for performance monitoring. Implement preventive maintenance based on analysis insights. Use analysis results for pilot training and mission planning optimization.

This comprehensive analysis provides valuable insights into the UAV flight performance and can be used for mission planning, pilot training, and system optimization. Regular analysis of flight data is recommended to maintain optimal performance and safety standards.